

Executing the Grey List: Interpretable Queries for Unfair-Term Detection in Terms of Service

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Abstract. The Annex of Directive 93/13/EEC lists seventeen kinds of contractual terms that may be regarded as unfair: the grey list. Automated detectors of unfair terms do not use it. They classify sentences with learned models whose decisions cite no legal ground. We ask what happens when the grey list is executed literally. Each clause of a contract, located by a human-validated theme, is described by a clause template stating who may do what, under which conditions, with what notice and what remedy; templates are extracted by a language model that sees neither the queries nor the reference labels. Grey-list items become queries over these templates, frozen with a recorded hash before any computation on held-out contracts, so that the result measures the list rather than our ability to fit it. Eleven of the seventeen items are expressible as structural conditions. On 17 held-out CLAUDETTE contracts, the frozen queries are precise on several items (0.61 to 0.83) but retrieve a third of the unfair sentences: their binary F_1 of 0.36 is below that of the clause theme alone (0.44) and of a fine-tuned legal encoder (0.70). Every flag cites the item, the template fields and the sentences that triggered it, and the error analysis locates where literal execution and the benchmark diverge. Queries, templates and code are released.

Keywords. unfair contract terms, terms of service, Directive 93/13, legal knowledge representation, explainable AI, information extraction

1. Introduction

Directive 93/13/EEC on unfair terms in consumer contracts [1] defines unfairness by a general test—a significant imbalance in the parties’ rights and obligations, contrary to good faith—and complements it with an Annex: an indicative and non-exhaustive list of seventeen kinds of terms, item (a) to item (q), that *may* be regarded as unfair. Practitioners call it the grey list. It is the closest thing European consumer law offers to an operational definition of the phenomenon, and it is written in terms of *who* may do *what* to *whom* under *which* conditions: a seller who may terminate a contract of indeterminate duration

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without reasonable notice item (*g*), alter the terms unilaterally without a valid reason specified in the contract item (*j*), or exclude liability for personal injury item (*a*).

Automated detectors of unfair terms do not use it. Since CLAUDETTE [2], the reference benchmark for online terms of service (ToS), the task has been framed as sentence classification: a learned model assigns one of eight unfairness categories to each sentence, and its decision cites no legal ground. Rationale-based extensions [3,4] attach explanations to predictions, but the explanations are learned too. Large language models prompted on the same task are neither more accurate nor more accountable [5,6].

This paper asks a simple question: **what happens if the grey list is executed literally?** We turn each item into a query over a structured representation of the contract: a *clause template* recording who acts, what is obliged, permitted or forbidden, and under which conditions, deadlines and remedies, attached to clauses identified by a human-validated thematic layer over the 50 CLAUDETTE contracts. Queries are written before any template is extracted and frozen before any computation on held-out contracts, so that the result is a measurement of the grey list and not of our ability to fit it. Three questions structure the paper, each with a hypothesis recorded in advance (Table 3). **RQ-A** (Section 6): which items can be expressed, and do the frozen queries retrieve the reference labels better than the clause theme alone? **RQ-B** (Section 7): what does this interpretability cost against learned text-only models? **RQ-C** (Section 8): what do the false positives reveal about the benchmark?

Because the Annex is indicative rather than binding, nothing guarantees that its items, executed literally, retrieve what the CLAUDETTE annotators marked as unfair. This is exactly what we measure. Our contributions are: (i) an executable, frozen encoding of the grey list, with an analysis of which items can be expressed as template conditions; (ii) a pre-registered, per-item evaluation against reference labels on held-out contracts, including the gain over the clause theme alone; (iii) a cost accounting of interpretability against text-only models trained on the same split; (iv) an error analysis in which every flag is traceable to the template fields and sentences that triggered it; and (v) the release of the queries with their hashes, the extracted templates and the code.

2. Background and Related Work

Unfair-term detection. CLAUDETTE [2,7] labels sentences of 50 ToS with eight categories—arbitration, unilateral change, content removal, jurisdiction, choice of law, limitation of liability, unilateral termination and contract by using—whose definitions are themselves derived from the Annex and from doctrine [8]. The corpus has been extended to several languages [9,10], included in LexGLUE [11], and used to compare fine-tuned encoders with prompted LLMs [5]. Related resources annotate clause validity in German consumer contracts [12] and potentially abusive clauses in Chilean ToS [13]. All of them predict; none of them cites the legal ground of a prediction.

Legal knowledge representation. Rule languages such as LegalRuleML [14] and contract specification languages such as Symboleo [15] formalise norms as deontic statements with parties, actions and conditions; they are designed for authoring and reasoning, not for being populated from natural-language contracts at scale. Deontic modality extraction [16,17] recovers obligations, permissions and prohibitions from text, with macro-F1 in the 0.6–0.8 range depending on the setting—good enough to be an input sig-

nal, not good enough to be a ground truth. Our templates sit between these two traditions: a fixed, shallow structure that lawyers can validate clause by clause and that queries can be written against.

Queries over contracts. ContractNLI [18] evaluates document-level hypotheses (“the receiving party may share confidential information with employees”) by learned entailment; CUAD [19] retrieves clause types for due diligence. Both pose the question we pose—does this contract contain a provision of this kind?—but answer it with a learned model. We answer it by executing an explicit condition and returning the template fields that satisfy it.

Explanation in AI and law. The field has long argued that legal decision support must explain in legal terms [20]. For unfair-term detection, explanations have so far been rationales selected by attention [3] or generated by a model [4]. A query that fires because a termination clause grants the provider a discretionary right without notice explains itself by construction, and the explanation can be wrong only in a way a lawyer can check.

3. The Executable Grey List: from the Annex to Frozen Queries

Table 1 lists the seventeen items of the Annex, the CLAUDETTE category each item most directly corresponds to, the clause theme under which such a term is found, and whether the item can be expressed as a condition over the clause templates defined below. Three groups emerge. *Structural items* describe a configuration of actor, modality, action and condition that a template can carry directly: items (a), (b), (d), (f), (g), (j), (k), (l), (m), (p) and (q). *Quantitative items* require a judgement of proportion or reasonableness that no template field encodes—“disproportionately high compensation” (e), “unreasonably early” (h)—and can be expressed only with an explicit, declared threshold. *Procedural items* depend on facts outside the text, such as whether the consumer had a real opportunity of becoming acquainted with the terms (i), and can be approximated at best by a proxy (acceptance by mere use, the CLAUDETTE category “contract by using”). Several categories of CLAUDETTE have no counterpart in the Annex: choice of law is not listed, and jurisdiction clauses are covered only through the general hindrance of legal action in (q). Conversely, items (c), (n) and (o) describe terms that are rare in ToS and absent from the reference categories. The grey list and the benchmark therefore overlap without coinciding, and the evaluation must be read item by item.

Clause templates. A template is a flat record attached to a clause (a maximal run of sentences sharing a theme). Its fields are fixed and few: the *actor* (provider, user, third party); the *modality* (obligation, permission, prohibition, power); the *action*, drawn from a closed inventory per theme (terminate, suspend, modify terms, modify service, change price, exclude liability, cap liability, license content, remove content, impose arbitration, waive class action, choose forum, choose law, . . .); the *object* or affected party; the *conditions* under which the action may be taken (for cause, for a specified reason, at discretion, none stated); the *notice or deadline* (a duration, “reasonable”, none, or not stated); and the *remedy* left to the other party (refund, right to cancel, compensation, none, not stated). Two conventions matter for the queries. *Not stated* is a value, distinct from *none*: a termination clause that says nothing about notice is not the same as one that excludes it, and grey-list items are sensitive to the difference. And a clause may carry several tem-

Table 1. The Annex of Directive 93/13/EEC (grey list), its correspondence with CLAUDETTE categories (A arbitration, CH unilateral change, CR content removal, J jurisdiction, LAW choice of law, LTD limitation of liability, TER unilateral termination, USE contract by using) and with clause themes, and whether the item is expressible as a template condition (S structural, Q quantitative with declared threshold, P procedural proxy, – not expressible from the text).

Item	Term that has the object or effect of...	CLAUDETTE	Clause theme	Expr.
(a)	excluding/limiting liability for death or personal injury	LTD	Limitation of liability	S
(b)	excluding/limiting the consumer’s legal rights in case of non-performance	LTD	Limitation of liability; Warranty	S
(c)	binding the consumer while the seller’s performance depends on his will alone	–	Preamble/scope	–
(d)	retaining sums paid when the consumer withdraws, without reciprocity	–	Fees and payment	S
(e)	requiring disproportionately high compensation from the consumer	–	Fees and payment	Q
(f)	dissolving the contract at the seller’s discretion without reciprocity	TER, CR	Termination; User content	S
(g)	terminating a contract of indeterminate duration without reasonable notice	TER	Termination	S
(h)	automatically extending a fixed-term contract with an unreasonably early deadline	–	Fees and payment	Q
(i)	binding the consumer to terms he had no real opportunity to know	USE	Preamble/scope	P
(j)	altering the terms unilaterally without a valid reason specified in the contract	CH	Modification of terms	S
(k)	altering unilaterally the characteristics of the product or service	CH	Modification of terms	S
(l)	determining the price at delivery or increasing it without a right to cancel	CH	Fees and payment	S
(m)	letting the seller decide conformity or interpret any term exclusively	–	Preamble/scope; Warranty	S
(n)	limiting the seller’s obligation to respect commitments of his agents	–	Third-party services	–
(o)	obliging the consumer to perform while the seller does not	–	–	–
(p)	transferring the seller’s rights and obligations without the consumer’s agreement	–	Preamble/scope	S
(q)	excluding or hindering legal action, imposing arbitration, restricting evidence	A, J	Arbitration and disputes	S

plates when it states several norms; queries are evaluated per template and aggregated per clause.

From an item to a query: item (g). The Annex targets terms “enabling the seller or supplier to terminate a contract of indeterminate duration without reasonable notice except where there are serious grounds for doing so”. Its constitutive elements map onto template fields: the provider acts (actor *provider*, modality *permission* or *power*, action *terminate* or *suspend*), the termination is tied to no ground (conditions $\in \{\text{at discretion, none stated}\}$) and no reasonable notice is given (notice $\in \{\text{none, not stated}\}$). The exception in the item itself (“serious grounds”) becomes a blocking condition: a template

whose condition is *for cause*, or that is stated as an exception to such a template, does not fire. Executing $Q(g)$ returns, for each hit, the template, the item and the sentences cited; Section 8 shows one.

Queries. In general, a query is a conjunction of conditions over template fields, optionally over several templates of the same document, that returns the templates satisfying it together with the sentences they were extracted from. The queries were drafted with a generative AI assistant from the text of the Annex and the CLAUDETTE category definitions, then reviewed and approved by the authors, before any template was extracted. They were revised once, on development contracts only: six action codes were added to the inventories, and two queries were extended, $Q(q)$ to contractual limitation periods and $Q(i)$ to acceptance deemed by continued use. The set was then frozen with a recorded hash, before any held-out template was extracted, and never edited afterwards. Every query cites the grey-list item it implements. Three further examples, in prose:

- $Q(j)$ — theme *modification of terms*; actor *provider*; action *modify terms*; conditions $\in \{\text{at discretion, none stated}\}$ and no valid reason specified; a variant $Q(j')$ additionally requires remedy $\in \{\text{none, not stated}\}$, that is, no right to cancel.
- $Q(a/b)$ — theme *limitation of liability*; action *exclude liability* or *cap liability*; object includes personal injury or gross negligence, or the exclusion is unbounded (no carve-out for mandatory law).
- $Q(q)$ — action $\in \{\text{impose arbitration, waive class action, restrict evidence, shift burden of proof, waive jury trial, limit the claim period}\}$ with modality *obligation* or *prohibition* on the user; or the provider holding the *power* to impose arbitration or choose the forum.

Queries come in three families: *presence* of a forbidden configuration (the four above); *absence* of an element the theme’s schema expects (a modification clause with no notification mechanism and no notice); and *composition* across clauses of the same document (a discretionary termination right of the provider with no symmetric termination right for the user). All are scored at the sentence level through the sentences their templates cite. A further composition pattern, combining an exclusion of liability, a warranty disclaimer and an indemnification duty of the user, is reported qualitatively only.

4. Materials: Corpus, Thematic Layer and Clause Templates

Corpus. CLAUDETTE comprises 50 online terms of service split into 9,414 sentences, of which 1,032 (11.0 %) carry at least one of 1,137 unfairness labels. The 17 held-out contracts contain 4,078 sentences, 440 of them labelled unfair.

Thematic layer. The clause themes we rely on come from a companion resource: a multi-label thematic annotation of the 9,414 sentences of CLAUDETTE by three trained annotators, with α -MASI 0.658 on the 20-theme taxonomy and 0.725 on its 11-theme consolidation, and a leave-one-annotator-out human ceiling of 0.871 accuracy (κ 0.859). Agreement measures follow standard practice [21,22,23]; disagreement is preserved rather than erased [24]. Every sentence of the 50 contracts carries a primary clause theme and optional secondary themes from a closed vocabulary of 20 themes, assigned independently by three trained annotators and resolved by a cascade (strict agreement 46.8 %, majority 48.3 %, human arbitration 4.9 %). Themes are not unfairness labels, but they

Table 2. Unfairness by clause theme on the 50 CLAUDETTE contracts (consensus theme; n sentences, share carrying a CLAUDETTE label, lift over the 11.0 % base rate). Themes with fewer than 20 sentences are omitted.

Theme	n	unfair	lift	Theme	n	unfair	lift
Governing law	164	54.3 %	4.95	Fees and payment	1,077	5.5 %	0.50
Modification of terms	392	49.0 %	4.47	Warranty disclaimer	613	5.4 %	0.49
Termination	422	43.4 %	3.96	Eligibility and account	761	4.2 %	0.38
Limitation of liability	604	37.2 %	3.40	Privacy and data	278	4.0 %	0.36
Arbitration and disputes	847	10.5 %	0.96	Licence and IP	1,225	3.4 %	0.31
User content	658	9.7 %	0.89	Acceptable use	911	3.4 %	0.31
Preamble and scope	677	9.5 %	0.86	Miscellaneous boilerplate	586	0.9 %	0.08
Third-party services	464	6.5 %	0.59	Meta, communications	184	0.0 %	0.00

are far from independent of them: Table 2 gives, for each theme, the share of sentences carrying a CLAUDETTE unfairness label. Four themes—governing law, modification of terms, termination and limitation of liability—concentrate the unfairness signal with lifts between 3.4 and 5.0 over the 11.0 % base rate; the framing themes (preamble, boilerplate, communications) are almost never unfair. This stratification matters twice: it is the reason why the theme alone is a strong baseline that any structured method must beat, and it is the reason why the 11-theme consolidation used downstream never merges themes across strata.

Template extraction. Templates are extracted automatically; clauses longer than eight sentences are split into windows of at most eight. A language model (Claude Opus 5) receives the clause sentences, the clause theme, the closed action inventory for that theme and the output schema, and returns zero or more templates, each citing the sentences that support it. It runs in an isolated context that contains neither the queries nor any reference label. Outputs are checked for schema conformance, structural validity (cited sentences exist, actions belong to the inventory) and semantic coherence; a lexical anchoring check then resets to *not stated* any condition, notice or remedy value that no cue in the cited sentences supports, a deliberately conservative correction. Templates were not reviewed by legal experts in this study (Section 9). On a development pilot of 100 clauses extracted three times, 98 % of outputs conformed to the schema, the decisive fields (actor, modality, action, condition, notice, remedy) agreed across runs with a mean Jaccard index of 0.81, and the sentences flagged by the frozen queries agreed across runs with Fleiss’ $\kappa = 0.86$. The held-out contracts were extracted twice. The pre-registered analysis uses the first pass: 1,087 clause windows, all schema-conformant, 99.9 % structurally and semantically valid, yielding 2,809 templates (2.58 per clause), with 9.7 % of windows carrying none. The second pass, evaluated with the same frozen analysis, serves as a sensitivity check: its decisive fields agree with those of the first with a mean Jaccard index of 0.77, and the sentences flagged by the queries agree with Fleiss’ $\kappa = 0.82$.

5. Pre-registered Evaluation Protocol

Split. The 50 contracts are split by document into 33 development contracts, used to build the field inventories, to debug the extraction and to revise the queries once, and 17 held-out contracts on which all numbers are reported. **Pre-registration.** The template schema

Table 3. Pre-registered questions, hypotheses and tests, recorded before any computation on the held-out contracts. Components requiring expert judgement are deferred to the follow-up study.

RQ	Hypothesis H1	Metric	Test
A	precision ≥ 0.60 on the mapped categories; recall ≥ 0.50 on LTD, TER and CH; F_1 gain over theme only on these three	per item and query: P, R, F_1	document-level bootstrap CI
B	loss in $F_1 \leq 0.05$ against the best text-only model (explanation sufficiency: deferred)	F_1 difference	non-inferiority, one-sided CI
C	$\geq 20\%$ of false positives are grounded flags missing from the reference (expert judgement: deferred)	false positives by item, theme, severity	descriptive only

and the queries were frozen, with a recorded hash and date, before the held-out templates were extracted; the extraction prompt is versioned and hashed as well. One change to the originally planned protocol was recorded at the same time: templates are used as extracted, without expert review, so the pipeline is evaluated end to end. **Reference.** A sentence is positive for a CLAUDETTE category if it carries that label in the original corpus; a query is scored against the category or categories its grey-list item maps to (Table 1), and, for items without a CLAUDETTE counterpart, against any unfairness label. **Metrics.** A flagged template marks only the sentences it cites. Per query: precision, recall and F_1 at the sentence level; per item: the same, aggregating the item’s queries; overall: micro- and macro-averaged over items, with document-level bootstrap intervals; and, for comparison with learned models, a binary *any unfairness* F_1 . **Baselines.** (i) *Theme only*: the best achievable retrieval from the clause theme alone, $P(\text{unfair} \mid \text{theme})$ estimated leave-one-document-out; (ii) a lexical floor, TF-IDF with logistic regression; (iii) a fine-tuned legal encoder [25]. Both learned models are trained on the development contracts and scored on the same held-out sentences; the difference in F_1 is the price paid for decisions that cite a legal ground. **Error analysis.** False positives and false negatives are compared with the reference labels and categorised by item, theme and CLAUDETTE severity, without expert adjudication.

6. Results A: Expressibility and Per-Item Retrieval

The theme alone is a strong baseline. On the full corpus, the theme-only predictor reaches an average precision of 0.404 against consensus themes and 0.418 against the arbitrated gold (base rate 0.110), that is, 3.7 times the chance level; the identity of the *combination* of themes carried by a clause reaches 0.589 in document-grouped cross-validation. Any structured method that does not clearly exceed these figures adds explanation but not signal.

Expressibility. Eleven of the seventeen items are structural and expressible without a threshold; two require a declared threshold; one is a procedural proxy; three cannot be expressed from the text of the clause (Table 1). The items that carry most of the CLAUDETTE labels—(a)/(b), (f)/(g), (j)/(k) and (q)—are all in the structural group.

Per-item retrieval. Table 4 reports, for each item scored on the held-out contracts, precision and recall against the mapped reference categories and the difference with the theme-only baseline. The queries are precise and sparse. Precision reaches 0.61 to 0.83

Table 4. Held-out retrieval per grey-list item (17 contracts, 4,078 sentences). n_+ : sentences carrying a mapped CLAUDETTE label; P, R, F_1 at sentence level with document-level bootstrap 95 % CI; theme only: P(label | theme) estimated leave-one-document-out; ΔF_1 : queries minus theme only. Items without a CLAUDETTE counterpart (flags: (d) 21; (e) 0; (h) 0; (m) 39; (p) 17) are not scored. Last row: micro-average over the structural items, which excludes the procedural proxy (i); macro- F_1 in parentheses.

Item	Ref.	n_+	Flagged	P	R	F_1 [95 % CI]	Theme	ΔF_1 [95 % CI]
(a)	LTD	132	4	0.50	0.02	0.03 [0.00; 0.08]	0.44	-0.41 [-0.50; -0.29]
(b)	LTD	132	19	0.79	0.11	0.20 [0.08; 0.31]	0.44	-0.24 [-0.40; -0.08]
(f)	TER,CR	140	12	0.83	0.07	0.13 [0.03; 0.26]	0.34	-0.21 [-0.33; -0.08]
(g)	TER	102	40	0.68	0.26	0.38 [0.27; 0.49]	0.36	+0.02 [-0.10; 0.13]
(i)	USE	44	37	0.43	0.36	0.40 [0.31; 0.46]	0.18	+0.21 [0.10; 0.30]
(j)	CH	88	42	0.62	0.30	0.40 [0.32; 0.50]	0.40	+0.00 [-0.10; 0.12]
(k)	CH	88	22	0.50	0.12	0.20 [0.08; 0.34]	0.40	-0.20 [-0.28; -0.11]
(l)	CH	88	18	0.61	0.12	0.21 [0.13; 0.29]	0.40	-0.19 [-0.33; -0.05]
(q)	A,J	45	103	0.17	0.38	0.23 [0.15; 0.32]	0.18	+0.05 [-0.04; 0.14]
Structural (8)				0.46	0.15	0.22 (macro 0.22)		-0.15

on five of the nine scored items ((b), (f), (g), (j), (l)), but recall never exceeds 0.38, and it stays at 0.30 on unilateral change, 0.26 on termination and 0.11 on limitation of liability. Over the eight structural items, that is, excluding the procedural proxy (i), micro-averaged precision is 0.46 and recall 0.15. The pre-registered hypothesis for RQ-A is therefore rejected: the recall threshold of 0.50 is missed on all three target categories. Against the theme alone, the queries are significantly better on one item only, (i), by +0.21 [0.10; 0.30] (Holm-adjusted permutation $p = 0.009$); they are indistinguishable on (g), (j) and (q), and significantly worse on (a), (b), (f) and (k). The gain on (i) is not robust to extraction noise: on the second extraction pass it shrinks to +0.14 [-0.01; 0.28] and loses significance (Holm-adjusted $p = 0.15$), while every other verdict is unchanged. Within an item, the query families behave as designed: the stricter variant Q(j'), which also requires the absence of a right to cancel, keeps precision (0.68) at half the recall (0.15); the absence query on modification clauses fires 8 times (precision 0.63); and the composition query behind (f), which requires that the document grant the user no symmetric termination right, is the most precise (0.83) and the sparsest (12 flags).

7. Results B: The Price of Interpretability

Table 5 compares the union of the frozen queries with three text-only systems on the same held-out sentences. The queries reach a binary F_1 of 0.36 [0.32; 0.42], with a precision of 0.41 (Wilson interval [0.36; 0.46]). The gap is -0.34 [-0.40; -0.28] against the fine-tuned legal encoder, -0.17 [-0.21; -0.12] against the lexical model and -0.07 [-0.12; -0.02] against the theme alone. Non-inferiority at the pre-registered margin of 0.05 is rejected in all three comparisons, and again on the second extraction pass, where the union reaches an F_1 of 0.34 [0.28; 0.41]. The encoder's misses are not a matter of threshold: 83 of its 101 false negatives receive an unfairness score below 0.1. The two approaches are not redundant, however. Of the 440 unfair sentences, the queries retrieve 145; the legal encoder finds 123 of them, 22 are found by the queries alone, and 79 are missed by both.

Table 5. The price of interpretability on the held-out sentences (any CLAUDETTE label, 440 positives among 4,078). All systems are scored by the same evaluation run; the learned models are trained on the 33 development contracts.

System	P	R	F_1 [95 % CI]	Flagged	Cites a legal ground
Theme only (leave-one-document-out)	0.42	0.46	0.44	489	no
TF-IDF + logistic regression	0.45	0.63	0.53 [0.49; 0.57]	614	no
Legal-BERT, fine-tuned	0.64	0.77	0.70 [0.65; 0.75]	527	no
Frozen queries (union)	0.41	0.33	0.36 [0.32; 0.42]	356	yes

8. Results C: When a Query Fires and the Benchmark Does Not

A flag reads as follows. In one held-out contract, the sentence “We may terminate your access to and use of the services, at our sole discretion, at any time and without notice to you” yields the template {actor: provider; modality: power; action: terminate; condition: at discretion; notice: none; remedy: not stated}; Q(*g*) fires, the flag cites item (*g*) and this sentence, and the reference labels the sentence as unilateral termination. Every flag returns the template fields that satisfied the query and the sentences they were extracted from, so each error can be traced to one of three sources: a mis-filled field, a query broader than its grey-list item, or a missing reference label for a clause that a literal reading of the Annex captures. In this study the attribution is automatic and partial. **Misses.** No missed sentence lacks a template altogether: every one lies in a clause for which templates were extracted. For unilateral change, most misses carry no template with the item’s action. For limitation of liability and termination, about half carry the right action but a field that blocks the query, most often an exclusion whose object names neither personal injury nor gross negligence (42 sentences for (*a*) and (*b*)), or a termination read as requiring cause (15 for (*g*)). **False positives.** Item (*q*) accounts for 86 of the 162 item-level false positives, all in dispute-resolution clauses that CLAUDETTE leaves unlabelled at the flagged sentence. Across scored items, 66 item-level false positives lie in a clause whose other sentences carry the mapped label, which points to the sentence-level granularity of the reference rather than to the query. Finally, the items without a CLAUDETTE counterpart fire 77 times: 39 flags for (*m*), none of them labelled, 17 for (*p*) and 21 for (*d*). Deciding whether such flags are grounded requires legal judgement; it is left to the expert study described in Section 9.

9. Discussion and Limitations

Three points follow. First, the grey list is executable, and executing it literally is precise but narrow. Where a query fires, it is right in most cases on several items, but the configurations the Annex describes cover a small part of what the CLAUDETTE annotators labelled: our queries retrieve a third of it, and a fine-tuned encoder more than twice as much. Second, the clause theme carries more retrievable signal than the templates add. What the templates contribute is *grounds*—the ability to say which item fired and why—rather than signal: the only significant gain, on one item, does not survive a second extraction pass. The price of these grounds is currently high, and the error analysis shows where it is paid: fields that block a query and actions missing from the templates, rather than absent templates. Third, the mismatch between the Annex and the benchmark is

itself a finding. Categories such as choice of law have no grey-list ground; items such as (*d*), (*m*) and (*p*) fire on the held-out contracts but have no CLAUDETTE counterpart, so a detector that executed the law faithfully would be penalised by the benchmark whenever it is right. **Limitations and future work.** The templates were not reviewed by legal experts, as this was not feasible within the time frame of this study. Our results therefore characterise the end-to-end pipeline, with extraction treated as a measured source of noise, and they do not establish the correctness of individual templates. The choice of extractor matters: on the development pilot, with the previous prompt version, a second model (gpt-6-astra) agreed with the one used here on the decisive fields with a mean Jaccard index of about 0.6. The held-out set was extracted twice only, so the stability of the held-out results rests on a single pair of runs. The reference labels are sentence-level while the Annex is clause-level; thresholds for quantitative items are declared, not derived; the held-out set is small and the intervals are wide; the queries reflect one reading of one legal system’s list. This work is part of an ongoing collaboration with legal experts: the legal co-authors helped design the architecture of the system and brought the legal practitioner’s perspective to it. The follow-up study addresses these limitations: independent double validation of the templates by two legal experts, with per-field agreement, which will separate extraction errors from query errors; re-extraction with pinned model versions and replication with an open-weight model; and an expert audit of the flagged sentences and of the explanations the queries return.

10. Conclusion and Availability

We have treated the grey list of Directive 93/13/EEC as what it is—a list of configurations of rights and obligations—and executed it as frozen queries over clause templates extracted automatically from the CLAUDETTE corpus. The result is a detector whose every decision cites the item it implements, and an item-by-item measurement of how much of the annotated unfairness a literal reading of the Annex retrieves: on held-out contracts, a precise but narrow third, below the clause theme alone and well below a learned encoder. The error analysis shows where literal execution fails and where the benchmark and the Annex part ways. Because the queries are explicit and frozen, the measurement can be repeated on any contract whose clauses carry themes, and revised item by item as the law or its reading changes. The queries with their hashes, the extracted templates, the code and the annotation platform used to build the thematic layer are released.

References

- [1] Council of the European Union. Council Directive 93/13/EEC of 5 April 1993 on unfair terms in consumer contracts; 1993. Official Journal of the European Communities, L 95, 21 April 1993.
- [2] Lippi M, Pařka P, Contissa G, Lagioia F, Micklitz HW, Sartor G, et al. CLAUDETTE: an automated detector of potentially unfair clauses in online terms of service. *Artificial Intelligence and Law*. 2019;27(2):117-39.
- [3] Ruggeri F, Lagioia F, Lippi M, Torrioni P. Detecting and explaining unfairness in consumer contracts through memory networks. *Artificial Intelligence and Law*. 2022;30(1):59-92.
- [4] Liepiņa R, Ruggeri F, Lagioia F, Lippi M, Drazewski K, Torrioni P. Explaining potentially unfair clauses to the consumer with the CLAUDETTE tool. In: *Proceedings of the Natural Language Processing Workshop (NLLP)*. vol. 2645 of CEUR Workshop Proceedings; 2020. .

- [5] Panarelli M, Galassi A, Lagioia F, Liepina R, Lippi M, Pałka P, et al. Is it worth using LLMs for unfair clause detection in terms of service? In: Proceedings of the 20th International Conference on Artificial Intelligence and Law (ICAIL); 2025. .
- [6] Frasheri M, Bakhtiarnia A, Esterle L, Iosifidis A. Are LLM-based methods good enough for detecting unfair terms of service?; 2024. ArXiv:2409.00077.
- [7] Lagioia F, Ruggeri F, Drazewski K, Lippi M, Micklitz HW, Torroni P, et al. Deep learning for detecting and explaining unfairness in consumer contracts. In: Legal Knowledge and Information Systems (JURIX 2019). vol. 322 of Frontiers in Artificial Intelligence and Applications; 2019. p. 43-52.
- [8] Micklitz HW, Pałka P, Panagis Y. The empire strikes back: digital control of unfair terms of online services. *Journal of Consumer Policy*. 2017;40:367-88.
- [9] Drawzeski K, Galassi A, Jabłonowska A, Lagioia F, Lippi M, Micklitz HW, et al. A corpus for multi-lingual analysis of online terms of service. In: Proceedings of the Natural Legal Language Processing Workshop (NLLP); 2021. p. 1-8.
- [10] Galassi A, Lagioia F, Jabłonowska A, Lippi M. Unfair clause detection in terms of service across multiple languages. *Artificial Intelligence and Law*. 2024.
- [11] Chalkidis I, Jana A, Hartung D, Bommarito M, Androutsopoulos I, Katz DM, et al. LexGLUE: a benchmark dataset for legal language understanding in English. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics; 2022. p. 4310-30.
- [12] Braun D, Matthes F. AGB-DE: a corpus for the automated legal assessment of clauses in German consumer contracts. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics; 2024. .
- [13] Löffler C, Martínez Freile A, Rey Pizarro T. Predicting potentially abusive clauses in Chilean terms of services with natural language processing. *Artificial Intelligence and Law*. 2025.
- [14] Athan T, Governatori G, Palmirani M, Paschke A, Wyner A. LegalRuleML: design principles and foundations. In: Reasoning Web. Web Logic Rules. vol. 9203 of Lecture Notes in Computer Science. Cham: Springer; 2015. p. 151-88.
- [15] Sharifi S, Parvizimosaed A, Amyot D, Logrippo L, Mylopoulos J. Symboleo: towards a specification language for legal contracts. In: Proceedings of the 28th IEEE International Requirements Engineering Conference (RE 2020). IEEE; 2020. p. 364-9.
- [16] Chalkidis I, Androutsopoulos I, Michos A. Obligation and prohibition extraction using hierarchical RNNs. In: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers). Melbourne: Association for Computational Linguistics; 2018. p. 254-9.
- [17] Sancheti A, Garimella A, Srinivasan BV, Rudinger R. Agent-specific deontic modality detection in legal language. In: Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing. Abu Dhabi: Association for Computational Linguistics; 2022. p. 11563-79.
- [18] Koreeda Y, Manning CD. ContractNLI: a dataset for document-level natural language inference for contracts. In: Findings of the Association for Computational Linguistics: EMNLP 2021. Punta Cana: Association for Computational Linguistics; 2021. p. 1907-19.
- [19] Hendrycks D, Burns C, Chen A, Ball S. CUAD: an expert-annotated NLP dataset for legal contract review. In: Advances in Neural Information Processing Systems, Datasets and Benchmarks Track; 2021. .
- [20] Atkinson K, Bench-Capon T, Bollegala D. Explanation in AI and law: past, present and future. *Artificial Intelligence*. 2020;289:103387.
- [21] Krippendorff K. Content Analysis: An Introduction to Its Methodology. 4th ed. Sage; 2018.
- [22] Passonneau RJ. Measuring agreement on set-valued items (MASI) for semantic and pragmatic annotation. In: Proceedings of the Fifth International Conference on Language Resources and Evaluation (LREC); 2006. .
- [23] Artstein R, Poesio M. Inter-coder agreement for computational linguistics. *Computational Linguistics*. 2008;34(4):555-96.
- [24] Braun D. I beg to differ: how disagreement is handled in the annotation of legal machine learning data sets. *Artificial Intelligence and Law*. 2024;32:839-62.
- [25] Chalkidis I, Fergadiotis M, Malakasiotis P, Aletras N, Androutsopoulos I. LEGAL-BERT: the muppets straight out of law school. In: Findings of the Association for Computational Linguistics: EMNLP 2020; 2020. p. 2898-904.